Analytical and Bioanalytical Chemistry (2026) 418:6861–6875
https://doi.org/10.1007/s00216-026-06683-z

RESEARCH PAPER

Comparative evaluation of pre-column derivatization strategies for PFAS analysis in paper-based food contact material using GC-MS

Charlotte Hauschildt[1,2] · **Christian Jung**[1] · **Hajo Haase**[2] · **Roland Becker**[1]

Received: 9 March 2026 / Revised: 24 June 2026 / Accepted: 14 July 2026 / Published online: 25 July 2026
© The Author(s) 2026

Abstract

The analysis of paper-based food contact materials (FCM) has gained increasing attention due to the widespread use of per- and polyfluoroalkyl substances (PFAS) and their known toxic effects and environmental persistence. As regulatory frameworks for PFAS in FCM remain limited, sensitive and reliable analytical methods are necessary to ensure food safety. While PFAS analysis is predominantly performed using liquid chromatography coupled to tandem mass spectrometry (LC-MS/MS), a systematic development of gas chromatography coupled to mass spectrometry (GC-MS)-based methods for multiple PFAS classes remains limited. In this study, two pre-column derivatization techniques were optimized and evaluated for GC-MS determination of perfluoroalkyl carboxylic acids (PFCA), fluorotelomer alcohols (FTOH), and fluorotelomer carboxylic acids (FTCA). Both derivatization methods enabled qualification and quantification of the targeted PFAS with comparable limits of detection in the low ng/mL range. N,N-Dimethylformamide dimethyl acetal (DMFDMA) derivatization proved to be more sensitive with a limit of detection down to 0.85 ng/mL. In addition, this derivatization approach was successfully automated within the GC-MS workflow, resulting in sample preparation times comparable to those of LC-MS-based methods. The application of this optimized method to paper-based FCM demonstrated its suitability for real world FCM analysis. Results of PFAS content per targeted PFAS of 26.6 to 261.4 ng/g highlight the potential of GC-MS analysis, particularly when combined with automated derivatization, as a viable alternative or complement to LC-MS for PFAS analysis in FCM.

Keywords PFAS · Gas chromatography-mass spectrometry · Derivatization · FCM · Automatization

Introduction

Per- and polyfluoroalkyl substances (PFAS), commonly known as "forever chemicals", have been in widespread industrial use since the 1940s [1–4], when polytetrafluoroethylene (PTFE) was first commercialized as Teflon® [5, 6]. In the following decades, the applications of PFAS expanded rapidly across the cosmetic, textile, electronics, and automotive industries, among others [7–9], due to their hydrophobic and oleophobic properties [1, 3, 8]. According to the *Organisation for Economic Co-operation and Development* (OECD), all compounds containing a fluorinated methyl ($-CF_3$) or methylene ($-CF_2-$) group are classified as PFAS, resulting in a variety of unique physicochemical properties [2, 10]. A total of 4730 PFAS can be assigned a CAS number recording to the OECD definition, while the overall group is estimated to comprise up to 10,000 different compounds [10, 11]. PFAS are nowadays associated with environmental persistence [12, 13], bioaccumulation [9, 14], adverse health effects [14–16], and toxicological risks [17, 18]. The first records of suspected toxic effects of PFAS were already made in the 1950 s [19, 20]. According to current knowledge, the assumed toxic effects range from carcinogenicity to neurotoxicity and systemic toxicity [14–17]. As a result of their toxic properties and ubiquitous occurrence [8, 11], some PFAS compounds have already been regulated. The regulatory groundwork for the EU was first laid by the REACH Regulation (EC No. 1907/2006) in 2006, which provided the EU's initial comprehensive framework for controlling chemical substances of industrial relevance, including PFAS [21]. Other EU regulations include the ban of perfluorooctanesulfonic acid (PFOS) since December

✉ Roland Becker
Roland.Becker@bam.de

[1] Department of Analytical Chemistry and Reference Materials, Bundesanstalt für Materialforschung und -prüfung (BAM), Berlin, Germany

[2] Department of Food Chemistry and Toxicology, Technische Universität Berlin, Berlin, Germany

2006 [22], and perfluorooctanoic acid (PFOA) since July 2020 [23]. As a representative example of numerous regulatory actions, the EU has set maximum levels for selected PFAS in certain foods of animal origin, including meat, fish, crustaceans, and eggs, since December 2022 under Commission Regulation (EU) 2022/2388 [24].

Food contact materials (FCM) are one of the pathways for human exposure to PFAS [25–28]. Direct contact with various food matrices, particularly at elevated temperatures, increases the migration of PFAS from FCM into food. Paper-based FCM are treated with PFAS to make them water- and oil-repellent, enabling their use as FCM [28–31]. PFAS have been detected in various paper-based FCM such as popcorn bags, baking paper, wrapping paper, burger boxes, and many more over the past decades [27, 32–36]. In addition, an increasing shift toward sustainability and the resulting transition from plastic to disposable and recyclable paper products has led to a significant increase in the consumption of paper-based FCM, making their analysis highly relevant to food safety [36–38]. To ensure food safety and compliance with these set limits in foodstuff, the development of analytical methods for the quantification of PFAS has been in focus for years. The current analytical standard for PFAS analysis is liquid chromatography coupled with (tandem) mass spectrometry (LC-MS/MS), which covers a broad spectrum of ionic PFAS such as perfluorocarboxylic acids (PFCA), perfluorosulfonic acid (PFSA), perfluoroalkyl phosphonic acid (PFPA), and nonionic PFAS with limit of detection (LOD) < 1 ppb [39–42]. Gas chromatography coupled with mass spectrometry (GC-MS) provides a less complex and more cost-efficient alternative [43, 44] that is well-suited for the analysis of nonionic PFAS such as FTOH and polyfluorinated sulfonamides (FOSA) [45–47]. A frequently cited limitation of GC-MS methods for PFAS detection is the requirement for derivatization, which is a time-consuming additional step during sample preparation [48–50]. For example, several derivatization approaches have been reported for the free carboxyl group of PFCA, including methylation [51–54], alkylation [13, 55, 56], silylation [50, 57], and amidation [43]. A further approach transformed PFCA into bis-(dimethylamino)-derivatives using N,N-dimethylformamide dimethyl acetal (DMFDMA) [58–60]. All these procedures have been applied to PFCA quantification in water [61, 62], soil [45, 52], foodstuff including milk and seafood, and food packaging [33, 60, 63]. In any case, GC-MS requires a quick and complete transformation into a derivative displaying improved chromatographic performance and analytical sensitivity.

Based on these approaches, this study optimized and compared two pre-column derivatization reagents for the simultaneous analysis of three PFAS classes, thereby investigating gas chromatography coupled to simple single quad mass spectrometry as a possibility for extending the analytical scope for these compounds [43, 48, 55, 60]. These methods enable the simultaneous determination of multiple PFCA, FTOH, and fluorotelomer carboxylic acids (FTCA), using the derivatization reagents DMFDMA and 2,3,4,5,6-pentafluorbenzylbromide (PFBBr). For the first time, a fully automated system for improved derivatization with DMFDMA has been developed. This approach integrates the derivatization step directly into the analysis, providing a fast, simple, and reproducible method. Following systematic optimization and comparative evaluation, the fully automated DMFDMA derivatization was implemented for the analysis of PFAS in paper-based FCM.

Materials and methods

Chemicals and standards

N,N-Dimethylformamide dimethyl acetal (DMFDMA, CAS 4637-24-5, > 98%) was purchased from Carl Roth GmbH + Co. KG (Karlsruhe, Germany). 2,3,4,5,6-pentafluorbenzylbromide (PFBBr, CAS 1765-40-8, 99%) was obtained from Merck/Sigma Aldrich (Darmstadt, Germany). All solvents used were of p.a. grade or Supra-Solv® grade. Methanol (MeOH), acetonitrile (ACN), toluene, and acetone were purchased from CHEMSOLUTE, Th. Geyer (Renningen, Germany). Tert-butyl methyl ether (MTBE) (for gas chromatography, > 99.8%, Supra-Solv®) was obtained from Merck. Potassium carbonate anhydrous (CAS 584-08-7) was supplied by J.T. Baker (Deventer, Netherlands) and sodium carbonate anhydrous (p.a., 99.8%, CAS 497-19-8) was purchased from CHEMSOLUTE, Th. Geyer. Ultrapure water (> 18.0 MΩ/cm at 25 °C) was produced with a Quantum® TEX Milli-Q from Merck. Further information regarding native and labelled standards used for GC-MS, as well as their abbreviations, can be found in the supplementary information (SI) Table S1.

Stock solutions

A standard stock solution of 16 different PFAS, containing PFCA (C4-C12, C14, C16, C18), FTOH (6:2, 8:2), and FTCA (8:2, 10:2) (details in Table S1), was prepared at a concentration of 50 µg/mL in respective solvent depending on the derivatization method. Subsequently, stock solutions were diluted to the appropriate concentrations for analytical measurements. For method validation a concentration range of 10 to 200 ng/mL in the respective solvent was used. To determine extraction recoveries a concentration of 25 µg/mL in acetonitrile was used.

Sample preparation

A total of eight paper-based FCM were collected for analysis. The materials were categorized based on their material properties: cardboard: $n=4$, bagasse: $n=3$, wheat pulp: $n=1$ (details in Fig. 5). The materials were provided in the framework of the EU project Screenfood (23IND13, ScreenFood). For validation purposes, spiked FCM analysis was performed using an optimized extraction method. Material spiking was carried out by even applying PFAS stock solution at a concentration of 25 µg/mL in acetonitrile to the selected FCM. Following the evaporation of acetonitrile, the dry spiked FCM was cut into pieces measuring 1 cm^2 with cleaned scissors rinsed with acetone and transferred to 15 mL PP Falcon™ Conical Centrifuge Tube (Thermo Fisher Scientific, Darmstadt, Germany). For material extraction, the respective raw material was cut into pieces of 1 cm^2 with cleaned scissors rinsed with acetone without spiking. Per extraction 1 g of raw material was used. PFAS extraction was done by sonication at room temperature with 5 mL of ACN for a total duration of 15 min. Then the extract was collected and filtered through disposable 0.45 µm regenerated cellulose syringe filters (CHROMAFIL Xtra RC, 0.45 µm, Macherey–Nagel, Düren, Germany) to remove any fine particles. The extraction was repeated in three consecutive extraction cycles (EC). To 200 µL aliquots of each EC, 10 µL of internal standard (IS) mix at a concentration of 500 ng/mL in acetonitrile was added after the extraction.

Derivatization with N,N-dimethylformamide dimethyl acetal

Derivatization was performed under optimized conditions by adding 10 µL of undiluted DMFDMA to a 200 µL aliquot of standard solution or acetonitrile sample extract. To ensure comparability, an automated sample preparation procedure was established using the autosampler MPS XL system (Gerstel, Mülheim a.d. Ruhr, Germany). In this procedure, 10 µL (2 × 5 µL) of DMFDMA was automatically added to a 200 µL aliquot, which was stored at 10 °C in the GC cooler tray. Following a 3 min incubation period with shaking at 300 rpm in the agitator module, the sample was injected into the cold injection system (CIS) of the GC. To program the automation script, the Gerstel Maestro1 software (version 3.5, Gerstel) was used.

Derivatization with 2,3,4,5,6-pentafluorobenzylbromide

For the derivatization of PFAS, 12.5 µL of undiluted PFBBr and 10 µL of freshly prepared 1 M K_2CO_3 in ultrapure H_2O were added to 1 mL aliquots of standard solution or acetone sample extract. The mixture was incubated at 65 °C for 60 min with gentle shaking at 400 rpm using the MHR 13 heated thermoshaker (HLC BioTech, Bovenden, Germany). Following incubation, the sample was cooled down to room temperature and subsequently filtered through a disposable 0.45 µm regenerated cellulose syringe filter. The sample was then automatically injected into the CIS of the GC-system for further analysis. All experimental parameters were optimized prior to analysis.

Instrumental analysis

GC-MS measurements were performed using a 7890 A gas chromatograph (Agilent Technologies, Waldbronn, Germany) equipped with a Gerstel Maestro autosampler (version 3.5, Gerstel, Mülheim a.d. Ruhr, Germany), a CIS (Gerstel), and a 5975 C linear quadrupole mass spectrometer (Agilent Technologies). The measurement system was operated using Mass Hunter Workstation software (version 10.0.368, Agilent Technologies) and integrated Maestro software (version 3.5, Gerstel) for automatic injections. The CIS program started with an equilibration time of 0.1 min at 30 °C and then a gradient of 12 °C/s to 275 °C, held for 5 min. In the injector, a 2 mm deactivated multiple-baffle liner was used. Injections were performed in split mode (1:5 or 1:2) or splitless mode, depending on the measurement method. The injected volume of all samples was 1 µL. The carrier gas was helium (5.0, Linde, Pullach, Germany) with a constant flow rate of 1.0 mL/min. Separation was achieved on a HP-5 MS capillary column (30 m × 0.25 mm id., 0.25 µm film thickness; Agilent J&W Scientific).

The oven program for the DMFDMA derivatives started at 40 °C (held for 3 min) and was ramped to 92.5 °C (held for 3 min) with 15 °C/min, then to 100 °C with 2 °C/min and finally to 320 °C with 35 °C/min (held for 3 min). The total duration of this optimized oven program is 22.54 min. The oven program for the PFBBr derivatives started at 40 °C (held for 1 min) and was ramped to 70 °C with 10 °C/min, followed by 30 °C/min to 95 °C (held for 1 min) and then to 100 °C with 5 °C/min. In the final steps, the temperature was ramped to 225 °C with 15 °C/min and then to 300 °C with a 30 °C/min (held for 2 min). The total duration of this optimized oven program is 19.66 min. The transfer line temperature was set to 320 °C for DMFDMA measurements and to 300 °C for PFBBr measurements. The mass spectrometer was operated in electron impact ionization (EI) mode at 70 eV, and the ion source was maintained at 230 °C. Mass spectra of the derivatized samples and extracts were acquired in full scan mode over a m/z range of 35 to 1000 for method development or in selected ion monitoring (SIM) mode for quantification. The data were qualitatively analyzed using *Mass Hunter Workstation Qualitative Analysis* software (version 10.0, Agilent Technologies) and quantitatively analyzed using *Mass Hunter Workstation Quantitative Analysis for GCMS and LCMS* software (version 10.1, Agilent

Technologies). For method calibration, eight equidistant calibration points in the range of 10 to 200 ng/mL per PFAS compound were measured in triplicates.

Quality assurance/quality control

PFAS-free consumables (PFAS-free septa, polypropylene GC vials, and septum-free syringes for automated injection) were used in all experiments. All solvents were tested for PFAS content (solvent blanks). Solvent blanks ($n = 3$) were determined using 200 mL solvent spiked with 25 ng of the labelled IS mixture (10 µL, concentration of 10 ng/mL in ACN). To ensure reproducibility and rule out system contamination, method and system blanks were measured for each analysis. Method blanks ($n = 3$) were performed injecting ACN after sample treatment. Spiked material experiments were conducted to ensure the quality of real sample analysis. To ensure repeatability, all spike experiments were conducted in triplicates. First, the number of extraction steps required for quantitative recovery of the analytes was tested. For this purpose, all analytes were spiked in a high concentration (1 µg/mL) onto selected FCM (details in the "Sample preparation" section). The limit of detection (LOD) and limit of quantification (LOQ) were obtained using the calibration curve method based on German standard DIN 32645:2008 [64] with the calibration points described in the proceeding section ($n = 3$). Details on this DIN method can be found in the SI.

Results and discussion

Derivatization of PFAS with N,N-dimethylformamide dimethyl acetal

DMFDMA has particularly been used for the derivatization of carboxylic acids [58–60]. The reaction with the carboxyl groups of PFCA yields bis-(dimethylamino)-derivatives (see Fig. 1a), which are more volatile and thermally stable, facilitating their efficient separation and detection by GC-MS [58, 60]. For the first time, this approach was extended to further PFAS classes to convert these into DMFDMA derivatives, enabling analysis of PFCA, FTOH, and FTCA simultaneously. The unsaturated fluorotelomer carboxylic acid (FTUCA) was also qualified using this method. FTUCA differs from FTCA in terms of HF-loss and has been reported as a possible intermediate degradation product [65–67]. The presence of FTUCA as a by-product in the FTCA standard was confirmed by LC-MS/MS analysis. The derivatization was performed straight forward by adding undiluted DMFDMA, followed by direct injection onto the CIS of the GC-MS system. The PFAS derivatives could be completely separated in less than 17 min under optimized GC-MS conditions (see Fig. 1b). To further simplify and standardize sample preparation, an automated sample preparation/derivatization was implemented using the Gerstel MPS XL Twister system. This step eliminates the need for manual derivatization, reducing the effort required for sample preparation. The automation results in a reduction of time required for sample preparation and an increase in reproducibility. To ensure reliable automated derivatization and thorough homogenization of the sample with the derivatization reagent, an incubation step with gentle shaking is integrated prior to injection.

The mass spectrum of derivatized 6:2 FTOH (see. Fig. 1c) shows fragment ions with m/z of 88, 388, 407, and 420 with high intensity. The base peak of m/z 88 corresponds to the molecular peak losing a methoxy group $[M\text{-}CH_3O]^+$. The highest fragment at m/z 420 corresponds to $[C_{11}H_{11}F_{13}NO]^+$. This fragment breaks further into the smaller fragment of m/z 407 $[C_{10}H_{10}F_{13}NO]$ losing a methyl group. Further fragmentation to m/z 388 with a mass difference of 19 indicates the loss of a fluorine atom. In case of derivatized 8:2 FTCA m/z 470 is the most abundant fragment (see Fig. 1d). This fragment results from the molecular ion losing a dimethylamino group $[M\text{-}N(CH_3)_2]^+$. The further fragmentation to m/z 451 indicates the loss of fluorine atom. The fragmentation pathway of 8:2 FTUCA is equivalent to that of 8:2 FTCA because both structures differ only in terms of HF loss (difference of 20 m/z). The fragmentation pathway of derivatized PFOA, exemplary for PFCA, 6:2 FTOH, and 8:2 FTCA were depicted in Fig. S1 in the SI. Representative mass spectra of PFOA, 6:2 FTOH, 8:2 FTCA, and 8:2 FTUCA can be found in Fig. S2. 8:2 and 10:2 FTUCA, which were detected as by-products of the FTCA standards, are not optimized or quantified further. However, the results demonstrate that FTUCA can be qualified using the described method, making it feasible to integrate this PFAS class into the analysis.

Method optimization

The derivatization conditions using DMFDMA were systematically optimized to ensure reliable and simultaneous detection of PFCA, FTOH, and FTCA. The first parameter investigated was the volume of the DMFDMA used for derivatization. The results are summarized for each PFAS class (see Fig. 2). To evaluate the effect of reagent volume on the signal intensity, reagent volumes ranging from 0.5 to 25 µL were tested. The resulting chromatographic peak areas were used to evaluate performance. For PFCA derivatives, the signal intensity was largely unaffected by the volume of DMFDMA (see Fig. 2a). In contrast, the signal intensity for derivatized FTOH depended strongly on the volume of DMFDMA within the range of 0.5–10 µL; less significant effects were observed at volumes between 10 and 25 µL. The derivatized 8:2 FTCA exhibited a

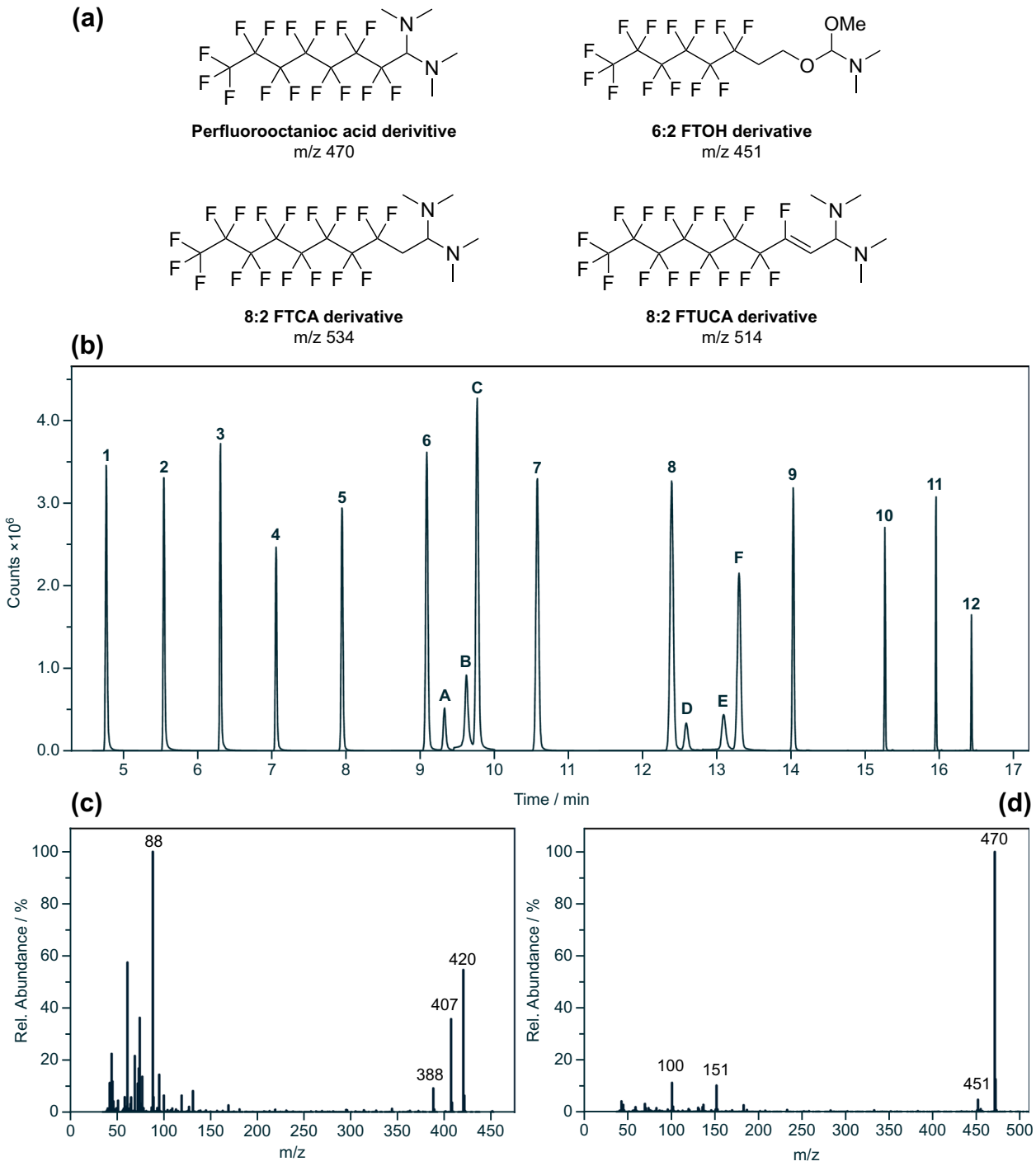


Fig. 1 **a** Structure of DMFDMA-derivatized, unfragmented molecules (molecular peaks) of PFOA, 6:2 FTOH, 8:2 FTCA, 8:2 FTUCA; **b** selected ion monitoring chromatogram of a gas chromatographic (GC) separation of 16 PFAS (1–12: PFCA C4-C12/C14/C16/C18; A+D: FTUCA (8:2, 10:2), B+E: FTOH (6:2, 8:2), C+F: FTCA (8:2, 10:2)) after derivatization with DMFDMA; **c** mass spectrum (EI, 70 eV) of 6:2 FTOH derivative; **d** mass spectrum (EI, 70 eV) of 8:2 FTCA derivative

saturating exponential curve that reached its plateau at approximately 10 µL of derivatization reagent (see Fig. 2c). As a result, a concentration of 10 µL DMFDMA was used for all further experiments.

Additionally, the influence of various solvents on derivatization efficiency was evaluated. The purpose of this optimization was to identify suitable solvents for subsequent extractions of PFAS-containing FCM. For this purpose,

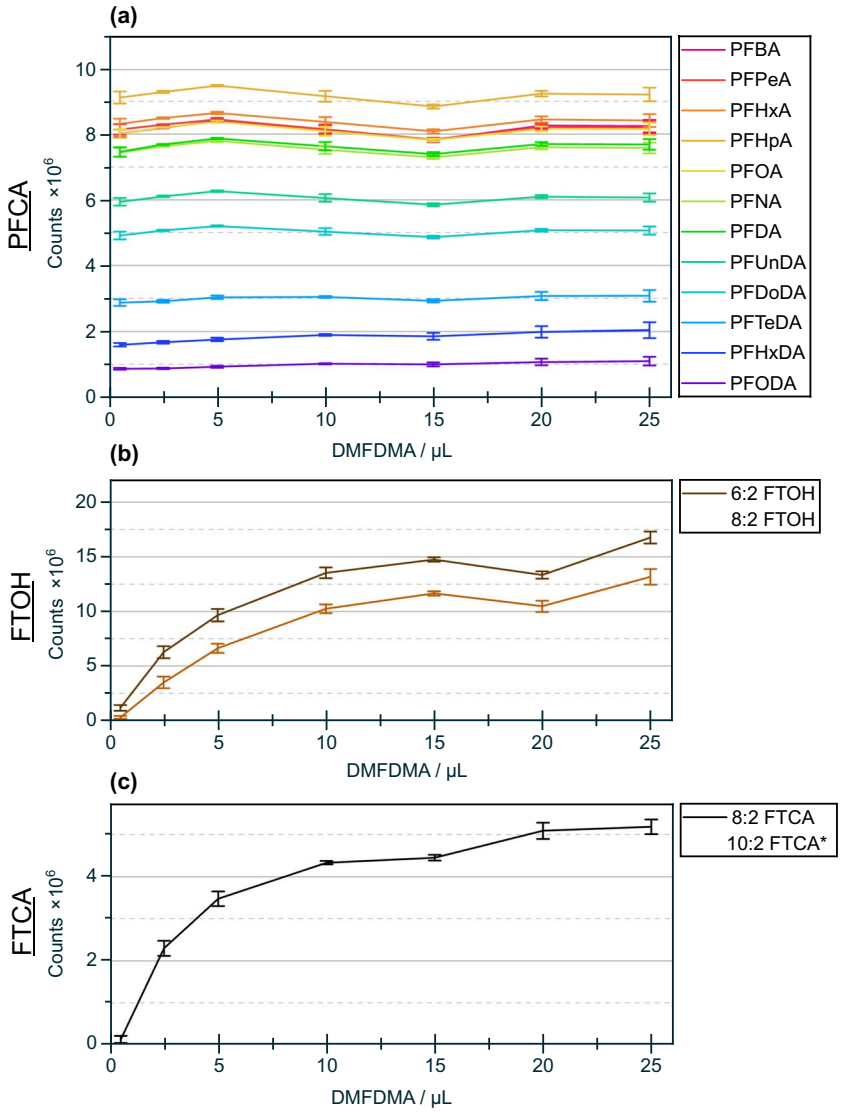


Fig. 2 Optimization of the DMFDMA volume for the derivatization of PFCA (**a**), FTOH (**b**), and FTCA (**c**). Evaluation of different volumes of DMFDMA (0.5–25 µL) in relation to a 200 µL aliquot of PFAS standard mix in acetonitrile, *10:2 FTCA was not analyzed, means and standard deviation ($n = 3$)

common solvents (MTBE, acetonitrile, ethyl acetate, acetone, and methanol) were compared with each other. Details are presented in Fig. S3. The results show that all solvents except methanol are suitable for the dilution of PFAS standards prior to DMFDMA derivatization. Methanol causes a rapid decrease in signal intensity within short time, resulting in inconsistent and unreliable measurements. For the analysis of FTOH neither ethyl acetate nor methanol led to sufficient derivatization. However, derivatization with DMFDMA works with MTBE, ACN, and acetone for all three PFAS classes. Furthermore, the stability of the derivatized PFAS was analyzed over a 24 h period. To assess this, the samples were stored at 15 °C in a GC cooler tray and measured hourly. As shown in Fig. S4, PFCA derivatives of DMFDMA remained largely stable with a slight decrease over the 24 h period. In contrast, FTOH derivatives are unstable over time. Intensity drops significantly over the 24 h period. Although derivatization for FTOH appears to work, the conditions would need to be optimized further to form stable products. The signal intensity of derivatized 8:2 FTCA increased slightly over the 24 h period. To enable reproducible quantification of FTOH despite the limited stability of the DMFDMA-derived product, an automated procedure with consistent timing throughout the entire sequence was developed (details of this automatization in the "Materials and methods" section).

Quantification

Following a successful optimization of the derivatization and chromatographic separation conditions, the LOD and LOQ as well as the coefficients of determination (R^2) were determined for each PFAS (see SI **DIN 32645:2008**). The LOD and LOQ were calculated using linear regression from eight measurement points in the range of 10–200 ng/mL ($n = 3$). Table 1 summarizes the chromatographic parameters,

Table 1 GC-MS parameter for all PFAS target analytes: retention time (RT), quantitative and qualitative ions, limit of detection (LOD) and limit of quantification (LOQ), and the coefficient of determination (R^2) for each target analyte after DMFDMA derivatization

PFAS	RT (min)	Quantitative ion (m/z)	Qualitative ions (m/z)	R^2	LOD (ng/mL)	LOQ (ng/mL)
PFBA	4.78	226	176, 107	0.9995	1.43	5.29
PFPeA	5.55	276	257, 226	0.9999	1.02	3.80
PFHxA	6.31	326	307, 276	0.9999	1.16	4.32
PFHpA	7.06	376	357, 326	0.9999	1.36	5.04
PFOA	7.94	426	407, 376	0.9999	1.20	4.46
PFNA	9.08	476	457, 426	0.9999	0.85	3.17
PFDA	10.57	526	507, 476	0.9999	1.08	4.04
PFUnDA	12.37	576	557, 526	0.9999	1.22	4.55
PFDoDA	14.01	626	607, 576	0.9988	4.49	15.85
PFTeDA	15.26	726	707, 676	0.9983	5.41	18.84
PFHxDA	15.94	826	807, 776	0.9984	5.37	18.70
PFODA	16.42	926	907, 876	0.9989	3.52	12.42
6:2 FTOH	9.62	420	407, 388	0.9994	4.87	17.51
8:2 FTOH	13.10	520	507, 488	0.9989	6.94	24.16
8:2 FTCA	9.77	470	451, 151	0.9982	9.07	30.65
10:2 FTCA	13.28	570	551, 151	0.9989	9.19	31.26

detection and quantification limits, and coefficient of determination that were used to validate the method and quantify PFAS in FCM.

The optimized method exhibited good linearity for all target analytes, with calibration ranges spanning from 10 to 200 ng/mL with R^2 ranging from 0.9982 to 0.9999. These R^2 values exceeding 0.9982 demonstrate that the linear regression model satisfactorily explains the observed data across all targeted analytes. The LODs for the PFCA derivatives ranged from 0.85 to 5.41 ng/mL, whereas LODs for FTOH and FTCA were between 6.94 and 9.19 ng/mL. Compared with literature data reporting PFAS concentrations in paper-based food FCM, the achieved LODs are within the concentration ranges previously reported [28, 68–70]. Therefore, the method presented here establishes a solid foundation for PFAS analysis in FCM. Compared to other FCM studies conducted using LC-MS/MS with LODs ranging down to 0.04 ng/g, subsequent refinements may further enhance its sensitivity [70, 71]. These improvements in sensitivity may be attained, for example by increasing quantification performance, e.g., by GC-MS/MS application for more sensitive trace-level analysis.

Derivatization of PFAS with 2,3,4,5,6-pentafluorbenzylbromide

PFAS derivatization with PFBBr was performed under optimized conditions by adding the undiluted derivatization reagent and 1 M K_2CO_3 to the sample aliquot, followed by an incubation step at 65 °C for 60 min. Although the resulting pentafluorobenzyl ester exhibit higher molecular weights (see Fig. 3a), they remain suitable for GC analysis due to their relatively low boiling points and high volatility [56]. Under optimized GC conditions, the separation of FTOH as well as PFCA, FTCA, and FTUCA derivatives was achieved in less than 14 min (see Fig. 3b).

Figure 3c and d show the representative spectra of 6:2 FTOH and 8:2 FTCA after PFBBr treatment, indicating that only FTCA is derivatized using PFBBr. The mass spectrum FTOH shows fragmentation ions of its non-derivatized form, which has a molecular mass of 364.10 g/mol. The most abundant peak of m/z 363 correlates to $[M\text{-}H]^+$. It is known from literature that the characteristic peaks of 6:2 FTOH are m/z 95 $[C_3H_5F_2O]^+$ as its base peak and m/z 69 $[CF_3]^+$ as the second highest [72]. The characteristic fragment of m/z 181, which corresponds to the pentafluorobenzylic cation, is not present after PFBBr treatment. That proves that no derivatization of FTOH with PFBBr has occurred. In contrast, the base peak of derivatized 8:2 FTCA is m/z 181 (see Fig. 3d). The fragment of m/z 658 results from the molecular ion losing the fluorinated benzyl alcohol group. The fragmentation pathway of 8:2 FTUCA is equivalent to that of 8:2 FTCA because both PFAS differ only in terms of HF loss (a difference of 20 m/z). The fragmentation pathways of derivatized PFOA, exemplary for PFCA, 6:2 FTOH, and 8:2 FTCA, were depicted Fig. S5 in the SI. Representative mass spectra of PFOA, 6:2 FTOH, 8:2 FTCA, and 8:2 FTUCA can be found in Fig. S6. FTUCA, which was detected as by-product of the FTCA standard, is not optimized or quantified further. However, the results demonstrate that FTUCA can be qualified using the described method, making it feasible to integrate this PFAS class into the analysis.

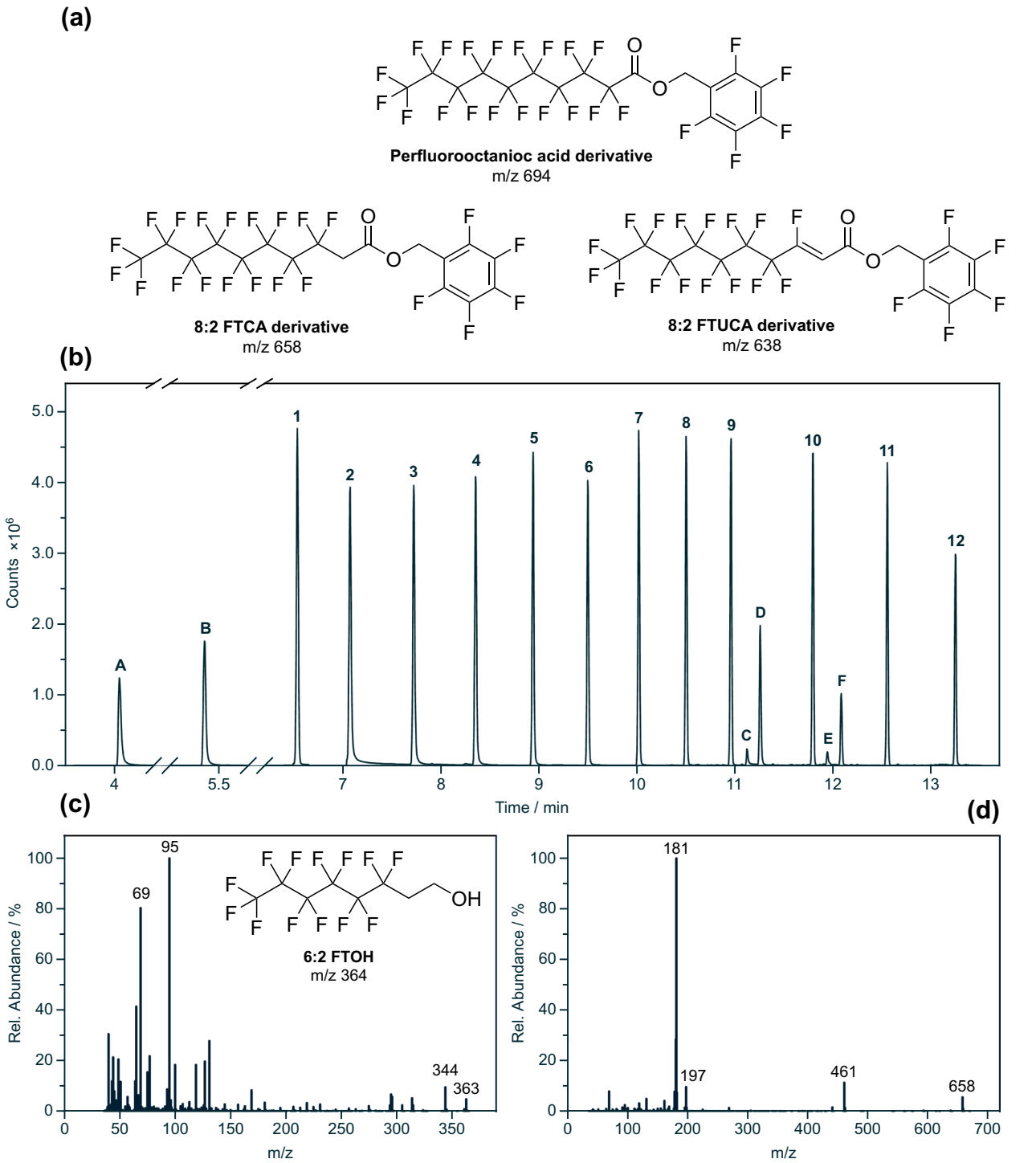


Fig. 3 **a** Structure of PFBBr-derivatized, unfragmented PFAS molecules (molecular peaks); **b** selected ion monitoring (SIM) chromatogram of a gas chromatographic (GC) separation of 16 PFAS (1–12: PFCA C4-C12/C14/C16/C18; A + B: 6:2 and 8:2 FTOH, C + E: 8:2 and 10:2 FTUCA, D + F: 8:2 and 10:2 FTCA) after derivatization with 2,3,4,5,6-pentafluorbenzylbromide (PFBBr); **c** mass spectrum (EI, 70 eV) of underivatized 6:2 FTOH; **d** mass spectrum (EI, 70 eV) of 8:2 FTCA derivative

Method optimization

Derivatization with PFBBr involves multiple parameters that can influence derivatization efficiency. To determine optimal conditions, these parameters were systematically optimized for PFCA, FTOH, and FTCA. Specifically, the effects of the amount of PFBBr, the choice and amount of carbonate used, the reaction temperature, and the incubation time were investigated. The results for the variation of PFBBr volume were summarized for each PFAS class (see Fig. 4). To assess the impact of PFBBr on the derivatization efficiency of PFCA, FTOH, and FTCA, the volume was varied from 2.5 to 15 µL. Derivatization efficiency was evaluated based on the chromatographic peak areas of the derivatized PFAS. For PFCA, the amount of PFBBr had a significant effect on derivatization efficiency. The optimal volume was found to be 12.5 µL of PFBBr (see Fig. 4a). In contrast, the experimental data show that the FTOH response does not depend on the volume of applied PFBBr in the range of 2.5–15 µL. This further confirms that FTOH does not react with PFBBr and is not derivatized under the conditions applied. For derivatized 8:2 FTCA higher signal intensities were observed at increased amounts of PFBBr. Consequently, 12.5 µL of PFBBr was used for all subsequent experiments.

In addition to the volume of PFBBr, further experimental parameters of the derivatization method were examined and optimized. These included the incubation temperature and duration, the type of added carbonate and its volume. The results can be seen in Fig. S7. These show that the incubation time between 30 and 90 min has almost no effect on the reaction. The incubation temperature, on the other hand, does affect the reaction. The signal intensity increases with temperature and asymptotically approaches a saturation level at elevated temperatures. Higher temperatures than 75 °C were not considered due to the solvent's vapor pressure. Using either Na_2CO_3 or K_2CO_3 leads to comparable results for

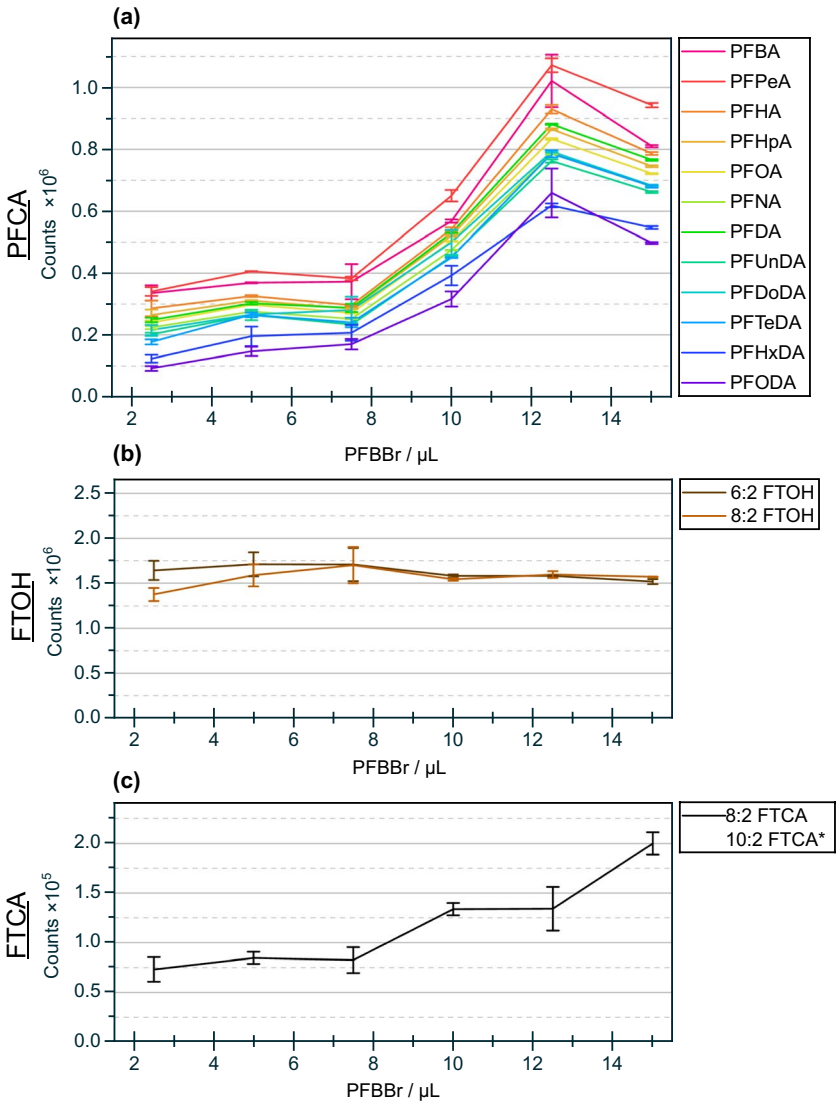


Fig. 4 Optimization of the derivatization reagent volume of PFBBr for PFCA (**a**), FTOH (**b**), and FTCA (**c**) analysis. Evaluation of different volumes (2.5–15 µL) of PFBBr in relation to a 1 mL sample of PFAS standard mix. *10:2 FTCA was not analyzed, means and standard deviation ($n = 3$)

all PFAS classes. Increasing the volume of K_2CO_3 solution slightly increased the signal intensity of derivatized PFCA and FTCA. A saturation level was reached at 10 µL added to a 1 mL aliquot. Therefore, the reaction parameters were set to 10 µL of 1 M K_2CO_3, 1 h incubation time, and 65 °C as the incubation temperature for all subsequent experiments. In addition, the stability of the PFBBr-derivatized PFAS was investigated. Therefore, each sample was analyzed at hourly intervals over a 24 h period. As shown in Fig. S8, PFCA and FTCA derivatives, as well as FTOH, remained largely stable throughout the 24 h observation period.

Quantification

After optimizing the derivatization parameters and the chromatographic separation conditions, the LOD, LOQ, and R^2 were evaluated for each PFAS compound using the calibration curve method mentioned above. Details can be found in SI. Table 2 shows each retention time (RT), quantitative and qualitative ions used, and the LOD, LOQ, and coefficient of determination.

The optimized method for PFBBr derivatization exhibited good linearity for all target analytes, with R^2 at least 0.9978 and calibration ranges spanning from 10 to 200 ng/mL. These R^2 values exceeding 0.9978 demonstrate that the linear regression model satisfactorily explains the observed data across all targeted analytes. The LODs for the PFCA derivatives ranged from 3.85 to 8.96 ng/mL, whereas LODs for FTOH and FTCA were between 5.49 and 13.19 ng/mL. The detection limits achieved for PFAS analysis with PFBBr are also within the concentration ranges previously reported in the literature for PFAS concentrations in paper-based FCM [28, 68–70]. Both methods establish a solid foundation for PFAS analysis in FCM. The LOQs achieved are sufficient for the determination of the most frequently detected PFAS. While LC-MS/MS-based methods have demonstrated higher sensitivity, partly attributable to the use of MS/MS detection and display as of yet a wider range of detectable PFAS, GC-MS is capable of quantifying PFCAs and FTOHs as the compound classes most relevant in FCM. Further method development could expand this range.

Derivatization method comparison

Both methods are suitable for analyzing PFCA, FTCA, and FTOH simultaneously in less than 20 min with comparable LOD ranges in the lower ng/mL range. Therefore, these methods largely meet the recently defined limit values of the EU regulation 2025/40 article 5(5), where limit values of 25 ppb for any PFAS as measured with targeted PFAS analysis has been defined for new FCM placed on the market [73]. A comparison of both methods shows that the main advantage of DMFDMA derivatization over PFBBr is its simplicity and time-efficiency as a one-step reaction without incubation time or temperature control. Automation further accelerates this process and ensures a high degree of reproducibility. Due to this automation, preparing samples using GC-MS is comparable in speed to analyzing PFAS using LC-MS. An advantage of derivatization using PFBBr is the stability of the resulting products and the fast chromatographic separation. Automatic sample preparation has not yet been developed because the addition of carbonate leads

Table 2 GC-MS parameter for all PFAS target analytes: retention time (RT), quantitative and qualitative ions, limit of detection (LOD) and limit of quantification (LOQ), and the coefficient of determination (R^2) according to each target analyte after PFBBr treatment

PFAS	RT (min)	Quantitative ion (m/z)	Qualitative ions (m/z)	R^2	LOD (ng/mL)	LOQ (ng/mL)
PFBA	6.43	394	181, 131	0.9985	8.30	29.15
PFPeA	6.90	444	181, 131	0.9983	8.96	31.37
PFHxA	7.55	494	181, 131	0.9984	8.69	30.47
PFHpA	8.19	544	181, 131	0.9984	8.50	29.83
PFOA	8.78	181	594, 131	0.9993	5.78	20.64
PFNA	9.35	181	644, 131	0.9996	3.85	13.91
PFDA	9.87	181	694, 131	0.9990	6.58	23.35
PFUnDA	10.36	181	744, 131	0.9990	6.84	24.23
PFDoDA	10.81	181	794, 131	0.9987	7.64	26.95
PFTeDA	11.97	181	894, 131	0.9995	4.69	16.87
PFHxDA	12.42	181	994, 131	0.9994	4.93	17.68
PFODA	13.11	181	169*, 131	0.9995	4.72	16.96
6:2 FTOH	3.91	95	363, 344	0.9992	5.88	20.98
8:2 FTOH	5.26	95	463, 444	0.9993	5.49	19.62
8:2 FTCA	11.12	658	181, 461	0.9978	13.19	43.99
10:2 FTCA	11.95	758	181, 561	n.d	n.d	n.d

*m/z 1094 is out of detection range. *n.d.*: not determined

to precipitation and tends to cause clogging of the injection needle without prior filtration. However, improved technology could make automation achievable and greatly shorten the sample preparation process by interleaving the individual samples. Although, the DMFDMA method was preferred for further usage due to its easy implementation and automation, and the lower LODs between 0.8 and 9.7 ng/mL.

Method application to paper-based food contact materials

PFAS are widely used in paper-based FCM due to their water- and oil-repellent properties, which are highly relevant for food industry applications [30, 74]. Thus, paper-based FCM were tested for PFAS. For this purpose, various materials were selected (cardboard: $n=4$, bagasse: $n=3$, wheat pulp: $n=1$). All materials were intended for direct contact with food at high temperatures. The optimized DMFDMA method described above was applied to analyze PFAS in eight FCM. Figure 5 shows the selected FCM made from cardboard, wheat pulp and bagasse divided into two categories of carboard materials and agricultural fiber materials.

To validate the developed DMFDMA derivatization method, spiked materials were initially tested. For this purpose, two different FCM, one of each category, were tested with a standard mix containing all three PFAS classes at 1 µg/mL. A labelled internal standard, added after the extraction, was used to compensate for instrument variability and drift effects. The extraction efficiency of each extraction cycle was tested in five consecutive steps ($n=3$). The results show that the extraction efficiency in three consecutive steps of a total of 5 steps tested averages 94.5% ± 2.5% for cardboard material and 88.9% ± 1.7% for bagasse material. The low RSD values obtained from triplicate analysis indicate good precision of the method, confirming that the analytical procedure provides consistent and reliable results across repeated measurements. Details on the extraction efficiency of each cycle and its relative standard deviation (RSD) for each targeted PFAS are shown in Tables S2 and S3. As a result, three consecutive steps were subsequently

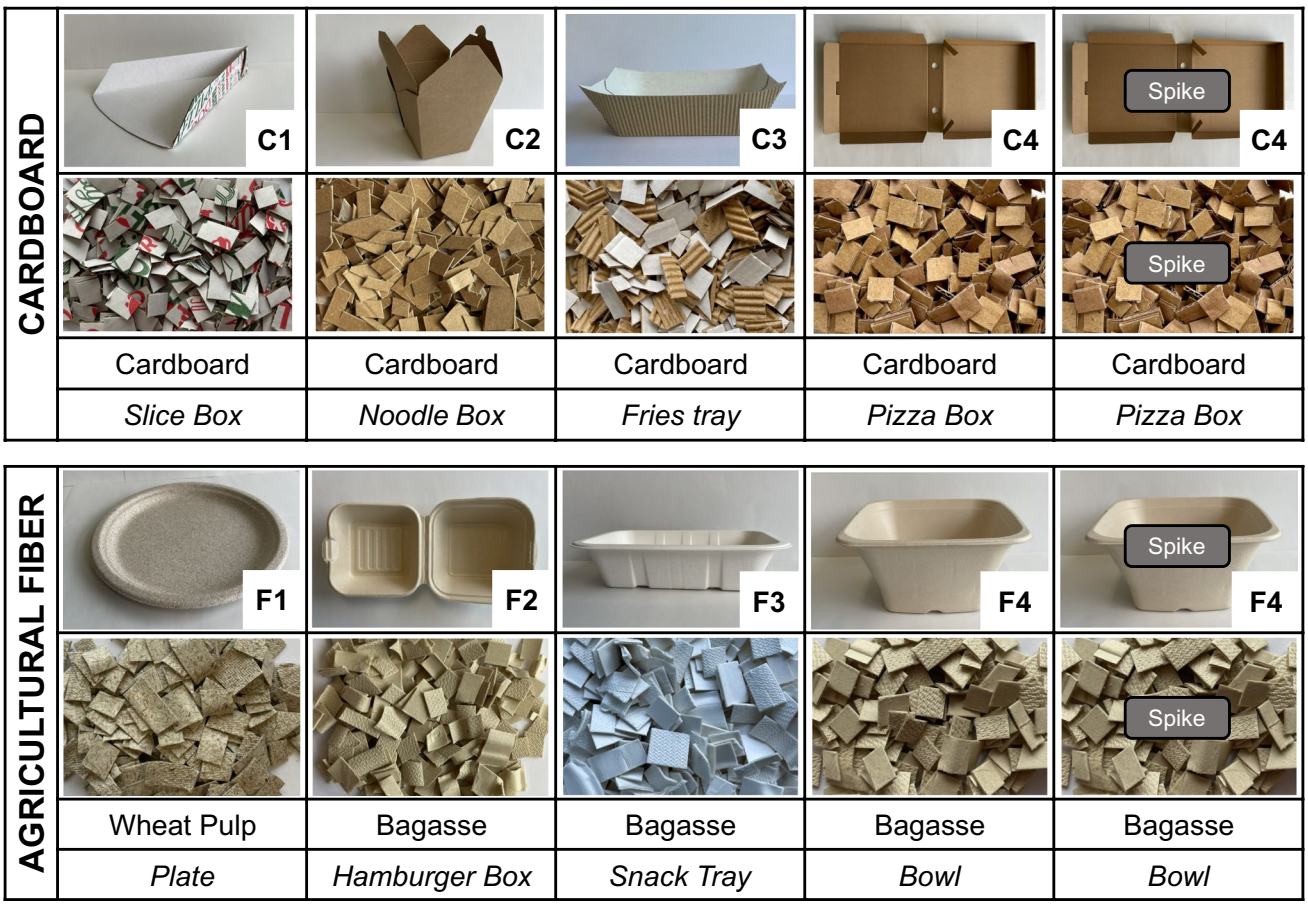


Fig. 5 Overview of analyzed paper-based food contact materials with details on material composition and intended use. Categorized in cardboard materials (C1–C4) and agricultural fiber materials (F1–F4). One material of each category was used for spiked experiments to investigate extraction and recovery rates (C4, F4)

determined to be sufficient for further experiments. In addition, the recovery of spiked PFAS standard mixture was determined in both material categories. The recovery rates for spiked cardboard samples extracted in three consecutive cycles range from 73.6 and 108.8% for PFCA, around 15% for FTOH, and between 21.3 and 42.3% for FTCA. For fiber materials, the recovery rates differ. For PFCA, the recovery in three consecutive cycles is between 26.6 and 73.1%, for FTOH between 130.9 and 144.2%, and for FTCA between 23.9 and 35.4% (details in Table S4). The overall precision of the recovery rates for the main extraction step (EC1) is very good, with RSD values below 10% for all analytes except 6:2 FTOH in the cardboard material with 14.3%, confirming the reliability and reproducibility of the extraction procedure. However, these results also indicate different material properties of each category, which may result from variations in manufacturing processes and PFAS application process used. This concept is further supported by the clearly distinguishable visual characteristics of the investigated FCM. Notably, the cardboard materials exhibit a markedly more porous surface structure, whereas the fiber-based materials appear considerably denser and, consequently, less absorbent. Furthermore, the food contact surfaces of all fiber-based materials appear significantly smoother, suggesting that they have undergone some form of surface treatment. In contrast, the surfaces of all cardboard materials are uniform in texture and appear to be untreated, except for partially printed areas. These observed differences in surface morphology may be attributable to the specific methods by which PFAS are incorporated into paper-based materials, such as coating, laminating, or sizing [30, 31, 75]. Therefore, further optimization of sample extraction specific to each material type may be required to make the analysis more effective.

This optimized method was applied to determine various PFAS in eight FCM made from cardboard and agricultural fibers. Three FCM containing PFCA and FTCA concentrations between 26.7 ng/g and 274.9 ng/g were detected (details in Tables S5 and S6). The highest amounts of PFAS were found in the snack tray (F3) made from bagasse. The predominant PFAS in the analyzed snack tray are 8:2 FTCA and PFDA. In addition, PFCA with a chain length of six to twelve could be detected. Also, in material C4 (pizza box made from cardboard) and F1 (plate made from wheat pulp), various PFCA were detected. In the analyzed pizza box, only elevated levels of short-chain PFCA between 31.1 and 62.6 ng/g were measured. FTOH were not detected in any of the analyzed materials, which could be due to the absence of FTOH in the selected materials. However, the very low recovery rates in cardboard and elevated recovery rates in fiber materials may also suggest the need for a further optimized extraction method. This could be likely related to the high volatility of FTOHs and the ubiquitous presence of FTOH in the air. In general, the PFAS concentrations determined in this study are consistent with values reported in recent literature [68, 76–79]. Most published data on PFAS analysis in FCM are based on LC-MS/MS methods. The GC-MS method presented here yields comparable results and demonstrates good reproducibility, as evidenced by low relative standard deviations, indicating its suitability as a complementary analytical approach. Furthermore, the limit values applicable from August 2026 for individual PFAS compounds, as well as the sum of PFAS measured by targeted PFAS analysis, as set out in EU Regulation 2025/40, are also largely met with the analysis presented [73]. From a green analytical chemistry perspective, the method presented here is characterized by low solvent consumption and comparably fast analysis times with sixteen analytes per run. The partly automated workflow further reduces the potential for human error and ensures ease of implementation in routine laboratory settings. Overall, this method offers a practical and resource-efficient alternative for the determination of PFAS in paper-based FCM.

Conclusion

This study optimized and validated two pre-column derivatization methods for GC-MS enabling the simultaneous determination of sixteen legacy and emerging PFAS including PFCA, FTOH, and FTCA. These methods are also shown to be suitable for the qualification of FTUCA but were not optimized for quantification. Key parameters affecting derivatization efficiency were systematically optimized, and an automated derivatization procedure was implemented for the DMFDMA-based approach, achieving sample preparation times comparable to LC-MS-based methods. To the best of our knowledge, this is the first report of an automated derivatization strategy for PFAS analysis by GC-MS. Both approaches achieved detection limits in the lower ng/mL range, which proved to be sufficient for the reliable analysis of paper-based FCM exhibiting elevated contamination levels. The application to FCM analysis demonstrated the suitability of this method for the rapid and sensitive analysis with GC-MS, with elevated PFAS levels detected in both cardboard and agricultural fiber-based materials, consistent with previous reports. Further improvements, for example coupling the developed derivatization methods with GC-MS/MS or expanding the PFAS range could enhance analytical sensitivity and selectivity even further. With continued optimization, this method may complement, and in selected applications provide an alternative to established LC-MS-based methods.

Supplementary Information The online version contains supplementary material available at https://doi.org/10.1007/s00216-026-06683-z.

Acknowledgements The present work has been supported by the project '23IND13 ScreenFood'. The project 23IND13 ScreenFood has received funding from the European Partnership on Metrology, co-financed from the European Union's Horizon Europe Research and Innovation Programme and by the Participating States.

Author contributions Conceptualization and methodology: C. Hauschildt, R. Becker. Experimental measurements: C. Hauschildt. Data analysis: C. Hauschildt, C. Jung. Preparation of the draft manuscript: C. Hauschildt. Review of the manuscript: C. Jung, H. Haase, R. Becker. Funding and acquisition: R. Becker. Supervision: H. Haase, R. Becker.

Funding Open Access funding enabled and organized by Projekt DEAL. The project 23IND13, ScreenFood has received funding from the European Partnership on Metrology, co-financed from the European Union's Horizon Europe Research and Innovation Programme and by the Participating States.

Data availability Data will be made available on request.

Code availability Not applicable.

Declarations

Ethics approval Not applicable.

Conflict of interest The authors declare no competing interests.

Open Access This article is licensed under a Creative Commons Attribution 4.0 International License, which permits use, sharing, adaptation, distribution and reproduction in any medium or format, as long as you give appropriate credit to the original author(s) and the source, provide a link to the Creative Commons licence, and indicate if changes were made. The images or other third party material in this article are included in the article's Creative Commons licence, unless indicated otherwise in a credit line to the material. If material is not included in the article's Creative Commons licence and your intended use is not permitted by statutory regulation or exceeds the permitted use, you will need to obtain permission directly from the copyright holder. To view a copy of this licence, visit http://creativecommons.org/licenses/by/4.0/.

References

1. Kissa E. Fluorinated surfactants and repellents: second edition, revised and expanded. 2nd ed. Marcel Dekker, Inc.; 2001.
2. Buck RC, Franklin J, Berger U, Conder JM, Cousins IT, de Voogt P, et al. Perfluoroalkyl and polyfluoroalkyl substances in the environment: terminology, classification, and origins. Integr Environ Assess Manag. 2011;74:513–41. https://doi.org/10.1002/ieam.258.
3. Richardson SD, Kimura SY. Water analysis: emerging contaminants and current issues. Anal Chem. 2016;881:546–82. https://doi.org/10.1021/acs.analchem.5b04493.
4. Mashima R. Per- and polyfluoroalkyl substances (PFAS): history, current concerns, and future outlook. Molecules. 2025. https://doi.org/10.3390/molecules30224415.
5. Plunkett RJ. Tetrafluoroethylene polymers. US2230654A. United States Patent Office; 1941.
6. Gaines L. Historical and current usage of per- and polyfluoroalkyl substances (PFAS): a literature review. Am J Ind Med. 2023;665:353–78. https://doi.org/10.1002/ajim.23362.
7. Glüge J, Scheringer M, Cousins IT, DeWitt JC, Goldenman G, Herzke D, et al. An overview of the uses of per- and polyfluoroalkyl substances (PFAS). Environ Sci Process Impacts. 2020;2212:2345–73. https://doi.org/10.1039/d0em00291g.
8. Key BD, Howell RD, Criddle CS. Fluorinated organics in the biosphere. Environ Sci Technol. 1997;319:2445–54. https://doi.org/10.1021/es961007c.
9. Giesy JP, Kannan K. Global distribution of perfluorooctane sulfonate in wildlife. Environ Sci Technol. 2001;357:1339–42. https://doi.org/10.1021/es001834k.
10. Organisation for Economic Co-operation and Development. Reconciling terminology of the universe of per- and polyfl uoroalkyl substances: recommendations and practical guidance. 2021.
11. Bundesministerium für Umwelt Naturschutz nukleare Sicherheit und Verbraucherschutz (BMUV). Per- und polyfluorierte Chemikalien (PFAS). 2024. https://www.bmuv.de/faqs/per-und-polyfluorierte-chemikalien-pfas. Accessed 17 Dec 2025.
12. Abunada Z, Alazaiza M, Bashir M. An overview of per- and polyfluoroalkyl substances (PFAS) in the environment: source, fate, risk and regulations. Water. 2020. https://doi.org/10.3390/w12123590.
13. Orata F, Quinete N, Wilken RD. Long chain perfluorinated alkyl acids derivatisation and identification in biota and abiota matrices using gas chromatography. Bull Environ Contam Toxicol. 2009;835:630–5. https://doi.org/10.1007/s00128-009-9756-y.
14. Sunderland EM, Hu XC, Dassuncao C, Tokranov AK, Wagner CC, Allen JG. A review of the pathways of human exposure to poly- and perfluoroalkyl substances (PFASs) and present understanding of health effects. J Expo Sci Environ Epidemiol. 2019;292:131–47. https://doi.org/10.1038/s41370-018-0094-1.
15. DeWitt JC. Toxicological effects of perfluoroalkyl and polyfluoroalkyl substances. 1st ed. In: Dietert RR, editor. Switzerland: Humana Cham; 2015.
16. Panieri E, Baralic K, Djukic-Cosic D, Buha Djordjevic A, Saso L. PFAS molecules: a major concern for the human health and the environment. Toxics. 2022. https://doi.org/10.3390/toxics10020044.
17. Stahl T, Mattern D, Brunn H. Toxicology of perfluorinated compounds. Environ Sci Eur. 2011. https://doi.org/10.1186/2190-4715-23-38.
18. Lindstrom AB, Strynar MJ, Libelo EL. Polyfluorinated compounds: past, present, and future. Environ Sci Technol. 2011;4519:7954–61. https://doi.org/10.1021/es2011622.
19. Sherwood RJ. The hazards of fluon (polytetrafluoroethylene). Trans Assoc Ind Med Off. 1955;5(1):10–2. https://doi.org/10.1093/occmed/5.1.10.
20. Robbins JJ, Ware RL. Pulmonary Edema from Teflon Fumes - Report of Case. New Engl J Med. 1964;2717:360–1. https://doi.org/10.1056/NEJM196408132710708.
21. The European Parliament and the Council of the European Union. Regulation (EC) No 1907/2006 of the European Parliament and of the Council of 18 December 2006 concerning the Registration, Evaluation, Authorisation and Restriction of Chemicals (REACH). 2006;L136:3–280.
22. The European Parliament and the Council of the European Union. Directive 2006/122/ECOF the European Parliament and of the Council of 12 December 2006 amending for the 30th time Council Directive 76/769/EEC on the approximation of the laws, regulations and administrative provisions of the Member States relating to restrictions on the marketing and use of certain dangerous substances and preparations (perfluorooctane sulfonates). 2006;L372:32–4.

23. The European Commission. Commission Delegated Regulation (EU) 2020/784 of 8 April 2020 amending Annex I to Regulation (EU) 2019/1021 of the European Parliament and of the Council as regards the listing of perfluorooctanoic acid (PFOA), its salts and PFOA-related compounds. 2020;L188:1–5.
24. The European Commission. Commission Regulation (EU) 2022/2388 of 7 December 2022 amending Regulation (EC) No 1881/2006 as Regards Maximum Levels of Perfluoroalkyl Substances in Certain Foodstuffs. 2022;L316:38–41.
25. Eze CG, Okeke ES, Nwankwo CE, Nyaruaba R, Anand U, Okoro OJ, et al. Emerging contaminants in food matrices: An overview of the occurrence, pathways, impacts and detection techniques of per- and polyfluoroalkyl substances. Toxicol Rep. 2024;12:436–47. https://doi.org/10.1016/j.toxrep.2024.03.012.
26. Begley TH, White K, Honigfort P, Twaroski ML, Neches R, Walker RA. Perfluorochemicals: potential sources of and migration from food packaging. Food Addit Contam. 2005;2210:1023–31. https://doi.org/10.1080/02652030500183474.
27. Schaider LA, Balan SA, Blum A, Andrews DQ, Strynar MJ, Dickinson ME, et al. Fluorinated compounds in U.S. fast food packaging. Environ Sci Technol Lett. 2017;43:105–11. https://doi.org/10.1021/acs.estlett.6b00435.
28. Ramirez Carnero A, Lestido-Cardama A, Vazquez Loureiro P, Barbosa-Pereira L, Rodriguez Bernaldo de Quiros A, Sendon R. Presence of perfluoroalkyl and polyfluoroalkyl substances (PFAS) in food contact materials (FCM) and its migration to food. Foods. 2021;107. https://doi.org/10.3390/foods10071443.
29. Peters R, Groeneveld I, Sanchez PL, Gebbink W, Gersen A, de Nijs M, et al. Review of analytical approaches for the identification of non-intentionally added substances in paper and board food contact materials. Trends Food Sci Technol. 2019;85:44–54. https://doi.org/10.1016/j.tifs.2018.12.010.
30. Trier X, Taxvig C, Rosenmai AK, Pedersen GA. PFAS in paper and board for food contact - options for risk management of poly- and perfluorinated substances. 2018. https://doi.org/10.6027/TN2017-573.
31. Gess JM, Rodriquez JM. The sizing of paper. Atlanta: Tappi; 2005.
32. Moreta C, Tena MT. Fast determination of perfluorocompounds in packaging by focused ultrasound solid-liquid extraction and liquid chromatography coupled to quadrupole-time of flight mass spectrometry. J Chromatogr A. 2013;1302:88–94. https://doi.org/10.1016/j.chroma.2013.06.024.
33. Liu X, Guo Z, Folk EEt, Roache NF. Determination of fluorotelomer alcohols in selected consumer products and preliminary investigation of their fate in the indoor environment. Chemosphere. 2015;129:81–6. https://doi.org/10.1016/j.chemosphere.2014.06.012.
34. Duenas-Mas MJ, Ballesteros-Gomez A, de Boer J. Determination of several PFAS groups in food packaging material from fast-food restaurants in France. Chemosphere. 2023;339:139734. https://doi.org/10.1016/j.chemosphere.2023.139734.
35. Trier X, Granby K, Christensen JH. Polyfluorinated surfactants (PFS) in paper and board coatings for food packaging. Environ Sci Pollut Res Int. 2011;187:1108–20. https://doi.org/10.1007/s11356-010-0439-3.
36. Barhoumi B, Sander SG, Tolosa I. A review on per- and polyfluorinated alkyl substances (PFASs) in microplastic and food-contact materials. Environ Res. 2022;206: 112595. https://doi.org/10.1016/j.envres.2021.112595.
37. Simantiris N. Single-use plastic or paper products? A dilemma that requires societal change. Clean Waste Syst. 2024. https://doi.org/10.1016/j.clwas.2023.100128.
38. Ciano S, Di Mario M, Goscinny S, Van Hoeck E. Towards less plastic in food contact materials: an in-depth overview of the Belgian market. Foods. 2023;1214. https://doi.org/10.3390/foods12142737.
39. Schiavone C, Portesi C. PFAS: a review of the state of the art, from legislation to analytical approaches and toxicological aspects for assessing contamination in food and environment and related risks. Appl Sci-Basel. 2023;13:1311. https://doi.org/10.3390/app13116696.
40. Jia S, Marques Dos Santos M, Li C, Snyder SA. Recent advances in mass spectrometry analytical techniques for per- and polyfluoroalkyl substances (PFAS). Anal Bioanal Chem. 2022;414(9):2795–807. https://doi.org/10.1007/s00216-022-03905-y.
41. Strozynska M, Schuhen K. Extraction and derivatization for perfluorocarboxylic acids in liquid and solid matrices: a review. Anal Sci Adv. 2021;27–8:343–53. https://doi.org/10.1002/ansa.202000089.
42. Suman TY, Kwak IS. Current understanding of human bioaccumulation patterns and health effects of exposure to perfluorooctane sulfonate (PFOS). J Hazard Mater. 2025;487: 137249. https://doi.org/10.1016/j.jhazmat.2025.137249.
43. Li Z, Sun H. Cost-effective detection of perfluoroalkyl carboxylic acids with gas chromatography: optimization of derivatization approaches and method validation. Int J Environ Res Public Health. 2019. https://doi.org/10.3390/ijerph17010100.
44. Santos FJ, Galceran MT. The application of gas chromatography to environmental analysis. TrAC Trends Anal Chem. 2002;21(9–10):672–85. https://doi.org/10.1016/s0165-9936(02)00813-0.
45. Bach C, Boiteux V, Hemard J, Colin A, Rosin C, Munoz JF, et al. Simultaneous determination of perfluoroalkyl iodides, perfluoroalkane sulfonamides, fluorotelomer alcohols, fluorotelomer iodides and fluorotelomer acrylates and methacrylates in water and sediments using solid-phase microextraction-gas chromatography/mass spectrometry. J Chromatogr A. 2016;1448:98–106. https://doi.org/10.1016/j.chroma.2016.04.025.
46. Morales-McDevitt ME, Becanova J, Blum A, Bruton TA, Vojta S, Woodward M, et al. The Air that we Breathe: Neutral and volatile PFAS in Indoor Air. Environ Sci Technol Lett. 2021;810:897–902. https://doi.org/10.1021/acs.estlett.1c00481.
47. Mok S, Lee S, Choi Y, Jeon J, Kim YH, Moon HB. Target and non-target analyses of neutral per- and polyfluoroalkyl substances from fluorochemical industries using GC-MS/MS and GC-TOF: insights on their environmental fate. Environ Int. 2023;182:108311. https://doi.org/10.1016/j.envint.2023.108311.
48. Shafique U, Schulze S, Slawik C, Kunz S, Paschke A, Schüürmann G. Gas chromatographic determination of perfluorocarboxylic acids in aqueous samples - a tutorial review. Anal Chim Acta. 2017;949:8–22. https://doi.org/10.1016/j.aca.2016.10.026.
49. Workman J Jr. A review of the latest separation science research in PFAS analysis. LCGC International; 2024. p. 12–8. https://doi.org/10.56530/lcgc.int.co7684c7.
50. Strozynska M, Schuhen K. Dispersive solid-phase extraction followed by triethylsilyl derivatization and gas chromatography mass spectrometry for perfluorocarboxylic acids determination in water samples. J Chromatogr A. 2019;1597:1–8. https://doi.org/10.1016/j.chroma.2019.03.008.
51. Belisle J, Hagen DF. A method for the determination of perfluorooctanoic acid in blood and other biological samples. Anal Biochem. 1980;101(2):369–76. https://doi.org/10.1016/0003-2697(80)90202-x.
52. Naile JE, Garrison AW, Avants JK, Washington JW. Isomers/enantiomers of perfluorocarboxylic acids: method development and detection in environmental samples. Chemosphere. 2016;144:1722–8. https://doi.org/10.1016/j.chemosphere.2015.10.075.
53. Ye R, Di Lorenzo RA, Clouthier JT, Young CJ, VandenBoer TC. A Rapid Derivatization for Quantitation of Perfluorinated Carboxylic Acids from Aqueous Matrices by Gas Chromatography-Mass Spectrometry. Anal Chem. 2023;9519:7648–55. https://doi.org/10.1021/acs.analchem.3c00593.

54. Huang G, Zhang B, Li Y, Zhu G, Jiang K. Determination of medium- and long-chain perfluorocarboxylic acids in water by catalytic methylation with trimethylsilyl diazomethane and gas chromatography - mass spectrometry analysis. Anal Chim Acta. 2025;1355: 344001. https://doi.org/10.1016/j.aca.2025.344001.
55. Ji YY, Cui ZY, Wang ZB, Cao YZ, Li XM, Li AD. Simultaneous determination of seven perfluoroalkyl carboxylic acids in water samples by 2,3,4,5,6-pentafluorobenzyl bromide derivatization and gas chromatography-mass spectrometry. Environ Pollut. 2020. https://doi.org/10.1016/j.envpol.2020.115043.
56. Tsikas D. Pentafluorobenzyl bromide-A versatile derivatization agent in chromatography and mass spectrometry: I. Analysis of inorganic anions and organophosphates. J Chromatogr B Analyt Technol Biomed Life Sci. 2017;1043:187–201. https://doi.org/10.1016/j.jchromb.2016.08.015.
57. Lv G, Wang L, Liu S, Li S. Determination of perfluorinated compounds in packaging materials and textiles using pressurized liquid extraction with gas chromatography-mass spectrometry. Anal Sci. 2009;253:425–9. https://doi.org/10.2116/analsci.25.425.
58. Elliott DE. Anomalous response off the flame ionization detector to perfluorinated carboxylic acids. J Chromatogr Sci. 1977;15(10):475–7. https://doi.org/10.1093/chromsci/15.10.475.
59. Stróżyńska M, Schuhen K. Derivatization of perfluorocarboxylic acids with N,N-dimethylformamide dimethylacetal prior to GC–MS analysis. Chromatographia. 2020;83(3):477–82. https://doi.org/10.1007/s10337-019-03850-6.
60. Ji Y, Cui Z, Li X, Wang Z, Zhang J, Li A. Simultaneous determination of nine perfluoroalkyl carboxylic acids by a series of amide acetals derivatization and gas chromatography tandem mass spectrometry. J Chromatogr A. 2020;1622: 461132. https://doi.org/10.1016/j.chroma.2020.461132.
61. Ayala-Cabrera JF, Contreras-Llin A, Moyano E, Santos FJ. A novel methodology for the determination of neutral perfluoroalkyl and polyfluoroalkyl substances in water by gas chromatography-atmospheric pressure photoionisation-high resolution mass spectrometry. Anal Chim Acta. 2020;1100:97–106. https://doi.org/10.1016/j.aca.2019.12.004.
62. Scott BF, Moody CA, Spencer C, Small JM, Muir DC, Mabury SA. Analysis for perfluorocarboxylic acids/anions in surface waters and precipitation using GC--MS and analysis of PFOA from large-volume samples. Environ Sci Technol. 2006;40(20):6405–10. https://doi.org/10.1021/es061131o.
63. Li M, Li P, Han Y, Han D, Yan H. Rapid and inexpensive nylon-66-filter solid-phase extraction followed by gas chromatography tandem mass spectrometry for analyzing perfluorinated carboxylic acids in milk. J Chromatogr A. 2022;1677: 463288. https://doi.org/10.1016/j.chroma.2022.463288.
64. Deutsches Institut für Normung e.V. DIN 32645:2008–11; Chemische Analytik – Nachweis-, Erfassungs- und Bestimmungsgrenze unter Wiederholbedingungen – Begriffe, Verfahren, Auswertung. 2008.
65. Wang N, Szostek B, Buck RC, Folsom PW, Sulecki LM, Gannon JT. 8-2 fluorotelomer alcohol aerobic soil biodegradation: pathways, metabolites, and metabolite yields. Chemosphere. 2009. https://doi.org/10.1016/j.chemosphere.2009.01.033.
66. Anumol T, Dagnino S, Vandervort DR, Snyder SA. Transformation of polyfluorinated compounds in natural waters by advanced oxidation processes. Chemosphere. 2016;144:1780–7. https://doi.org/10.1016/j.chemosphere.2015.10.070.
67. Butt CM, Muir DC, Mabury SA. Elucidating the pathways of poly- and perfluorinated acid formation in rainbow trout. Environ Sci Technol. 2010;4413:4973–80. https://doi.org/10.1021/es100702a.
68. Sapozhnikova Y, Taylor RB, Bedi M, Ng C. Assessing per- and polyfluoroalkyl substances in globally sourced food packaging. Chemosphere. 2023;337: 139381. https://doi.org/10.1016/j.chemosphere.2023.139381.
69. Encarnacion EKP, Alcantara AC, Armario HE, Alejandro WP, Zhan Z, Sun Z, et al. Preliminary screening of per- and polyfluoroalkyl substances (PFAS) in Philippine fast food packaging using liquid chromatography–mass spectrometry (LC-MS). Curr Res Nutr Food Sci J. 2024;121:423–36. https://doi.org/10.12944/crnfsj.12.1.34.
70. Siao P, Tseng SH, Chen CY. Determination of perfluoroalkyl substances in food packaging in Taiwan using ultrasonic extraction and ultra-performance liquid chromatography/tandem mass spectrometry. J Food Drug Anal. 2022;30(1):11–25. https://doi.org/10.38212/2224-6614.3397.
71. Hoang AQ, Tran TL, Tuyen LH, Nguyen TAH, Pham DM, Nguyen TC, et al. Perfluoroalkyl substances in food contact materials: preliminary investigation in Vietnam and global comparison. Environ Sci Pollut Res Int. 2023;3047:104181–93. https://doi.org/10.1007/s11356-023-29746-5.
72. Robbins ZG, Liu X, Schumacher BA, Smeltz MG, Liberatore HK. Method development for thermal desorption-gas chromatography-tandem mass spectrometry (TD-GC-MS/MS) analysis of trace level fluorotelomer alcohols emitted from consumer products. J Chromatogr A. 2023;1705: 464204. https://doi.org/10.1016/j.chroma.2023.464204.
73. Regulation (EU) 2025/40 of the European Parlianment and of the Council of 19 December 2024 on packaging and packaging waste, amending Regulation (EU) 2019/1020 and Directive (EU) 2019/904, and repealing Directive 94/62/EC. L series. 2025;1–124.
74. Borg D, Ivarsson J. Analysis of PFASs and TOF in products. TemaNord, editor. Nordic Council of Ministers, Nordic Council of Ministers Secretariat; 2017.
75. Guzman-Puyol S. Fluorinated compounds in paper and paperboard based food packaging materials. NPJ Sci Food. 2024;81:82. https://doi.org/10.1038/s41538-024-00326-2.
76. Zabaleta I, Blanco-Zubiaguirre L, Baharli EN, Olivares M, Prieto A, Zuloaga O, et al. Occurrence of per- and polyfluorinated compounds in paper and board packaging materials and migration to food simulants and foodstuffs. Food Chem. 2020;321: 126746. https://doi.org/10.1016/j.foodchem.2020.126746.
77. Zabaleta I, Negreira N, Bizkarguenaga E, Prieto A, Covaci A, Zuloaga O. Screening and identification of per- and polyfluoroalkyl substances in microwave popcorn bags. Food Chem. 2017;230:497–506. https://doi.org/10.1016/j.foodchem.2017.03.074.
78. Kotthoff M, Muller J, Jurling H, Schlummer M, Fiedler D. Perfluoroalkyl and polyfluoroalkyl substances in consumer products. Environ Sci Pollut Res Int. 2015;22(19):14546–59. https://doi.org/10.1007/s11356-015-4202-7.
79. Moreta C, Tena MT. Determination of perfluorinated alkyl acids in corn, popcorn and popcorn bags before and after cooking by focused ultrasound solid-liquid extraction, liquid chromatography and quadrupole-time of flight mass spectrometry. J Chromatogr A. 2014;1355:211–8. https://doi.org/10.1016/j.chroma.2014.06.018.

Publisher's Note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.